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HANZO INDUSTRIES

# Multi-Level Security and Cross-Domain Solutions via Homomorphic Policy Evaluation and Zero-Trust Fabric

TECHNICAL WHITE PAPER

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Antje Worring, Zach Kelling — Hanzo Team

Hanzo Industries Lux Industries Zoo Labs Foundation
research@hanzo.ai

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**Abstract**

We present a cross-domain solution (CDS) architecture where data transfer policies are evaluated on encrypted content using fully homomorphic encryption (FHE), ensuring the cross-domain guard never observes plaintext. The architecture integrates four components: (1) a zero-trust overlay network with identity-first micro-segmentation enforcing mandatory access controls at every hop; (2) an FHE policy evaluation engine (ThresholdVM) performing content inspection on ciphertexts using BFV/BGV/CKKS/TFHE schemes with GPU-accelerated NTT operations; (3) a key management system with Shamir-based root key distribution, per-organization encryption isolation, and HIPAA/SOX/SEC compliance controls; and (4) an organization-scoped identity and access management layer deriving security labels from JWT claims and enforcing them at the API gateway. The result is a multi-level security (MLS) architecture where data flows between security domains are cryptographically mediated, no single component has access to both plaintext and policy decisions, and all cross-domain transfers produce immutable audit records anchored to a post-quantum blockchain.

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# 1 Introduction

Cross-domain solutions enable controlled information sharing between networks operating at different security classifications. Traditional CDS architectures rely on trusted guards that inspect plaintext content against release policies—creating a high-value target where a single compromise exposes all cross-domain traffic.

This paper presents a fundamentally different approach: the guard evaluates policies on *encrypted* content using fully homomorphic encryption. The guard learns only the Boolean policy decision (release/deny), never the content itself. This eliminates the single point of trust while maintaining mandatory policy enforcement.

## 1.1 Problem Statement

Naval operations require information sharing across multiple security domains:

- **Unclassified** ↔ **Secret**: Intelligence products sanitized for coalition partners.
- **Secret** ↔ **Top Secret/SCI**: Analyst workstations accessing compartmented data.
- **US** ↔ **Coalition**: Multinational operations requiring controlled release.
- **Operational** ↔ **Administrative**: Preventing data spillage between networks.

Current CDS solutions (ISSE Guard, Radiant Mercury, High Speed Guard) process plaintext, requiring the guard itself to be trusted at the highest classification level. A compromised guard exposes all cross-domain traffic.

## 1.2 Our Approach

1. Data encrypted at source under FHE public key.
2. Guard evaluates content policy homomorphically on ciphertext.
3. Encrypted Boolean result threshold-decrypted by MPC committee.
4. If approved, data re-encrypted for destination domain.
5. Guard never sees plaintext; committee never sees policy result.

# 2 Architecture Overview

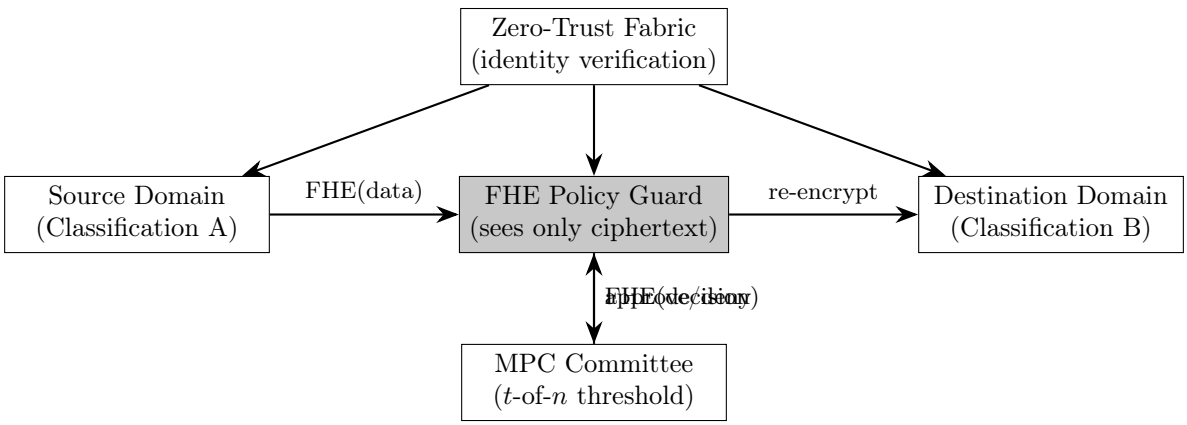


Figure 1: FHE-based cross-domain solution architecture.

### 2.1 Security Invariants

1. **Guard blindness**: The policy guard processes only ciphertexts. It cannot distinguish encrypted classified data from encrypted noise.
2. **Committee separation**: MPC committee members see decryption shares, never the full policy result or the data. Threshold ($t$-of-$n$) prevents minority collusion.
3. **Mandatory enforcement**: Zero-trust fabric verifies cryptographic identity at every hop. Security labels derived from JWT claims cannot be forged.
4. **Audit immutability**: Every cross-domain transfer produces a blockchain-anchored receipt with PQ signatures.

## 3 Zero-Trust Network Fabric

### 3.1 Identity-First Access Control

Hanzo ZT provides the network underlay with mandatory access controls:

- **Cryptographic identity**: Every endpoint possesses a unique X.509 certificate with HSM-backed private key (PKCS#11, YubiHSM).
- **No implicit trust**: Network position (IP address, VLAN, subnet) confers zero access. Every connection requires identity verification.
- **Micro-segmentation**: Policies define which identities can communicate. Cross-domain traffic flows only through designated guard endpoints.
- **Policy enforcement**: Applied at every hop in the overlay mesh, not just at perimeter firewalls.

### 3.2 Security Label Propagation

Security labels are derived from the IAM identity system:

Listing 1: Security label derivation from JWT claims.

```
// Gateway extracts identity from JWT
X-Org-Id:    "navy-secret"     // Organization = domain
X-User-Id:   "analyst-0042"    // Subject identity
X-User-Email: "j.doe@navy.mil" // Attribution

// Gateway enforces: only "navy-secret" org members
// can reach services in "secret" namespace
// Cross-domain requires FHE guard intermediary
```

The API gateway strips all client-supplied identity headers and derives them exclusively from cryptographically verified JWT claims. This prevents label forgery.

### 3.3 Multi-Organization Isolation

The platform supports 4+ organizations with complete cryptographic isolation:

| Organization | Root Key | Isolation Mechanism |
|---|---|---|
| hanzo | Separate AES-256 | KMS per-org encryption |
| lux | Separate AES-256 | Separate key hierarchy |
| zoo | Separate AES-256 | Org-scoped IAM policies |
| pars | Separate AES-256 | Independent audit logs |

Table 1: Per-organization cryptographic isolation.

Each organization has its own root encryption key, IAM policies, KMS secrets, and audit trail. Cross-organization data access requires explicit policy and traverses the FHE guard.

# 4 FHE Policy Evaluation

## 4.1 ThresholdVM: Computation on Encrypted Data

ThresholdVM operates on the Lux T-Chain, providing FHE as a native virtual machine:

| Scheme | Domain | CDS Application |
|---|---|---|
| BFV | Integer | Keyword matching on encrypted text |
| BGV | Modular | Classification marker detection |
| CKKS | Approx. real | ML-based content classification |
| TFHE | Boolean | Arbitrary policy Boolean circuits |

Table 2: FHE schemes mapped to CDS operations.

## 4.2 Policy Evaluation Flow

**Algorithm 1** FHE Cross-Domain Policy Evaluation

**Source** encrypts data: $c \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{data})$
**Source** encrypts metadata: $m \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{labels})$
**Source** sends $(c, m, \text{dest_domain})$ to Guard

**Guard** evaluates policy homomorphically:
  $r \leftarrow \text{FHE.Eval}(\text{policy_circuit}, c, m, \text{dest_domain})$
  (Guard sees only ciphertexts; $r$ is an encrypted Boolean)

**Guard** sends $r$ to MPC committee for threshold decryption
**for** each committee member $i = 1, \ldots, n$ **do**
  $d_i \leftarrow \text{ThresholdDecrypt}(\text{sk}_i, r)$
**end for**
decision $\leftarrow \text{Reconstruct}(d_1, \ldots, d_t)$  ($t$ shares suffice)

**if** decision = APPROVE **then**
  Re-encrypt data for destination domain key
  Forward to destination via zero-trust fabric
  Anchor audit receipt to blockchain (ML-DSA-65 signed)
**else**
  Log denial (encrypted), destroy ciphertext
**end if**

## 4.3 GPU-Accelerated FHE Operations

LuxCPP provides Metal and CUDA kernels for FHE primitives:

- **NTT/iNTT**: Number Theoretic Transform for polynomial multiplication in $\mathbb{Z}_q[X]/(X^n + 1)$.
- **TFHE bootstrap**: Programmable bootstrap for refreshing ciphertext noise.
- **Blind rotate**: Key operation in TFHE gate evaluation.
- **Batch operations**: Parallel ciphertext processing across GPU threads.

GPU acceleration reduces FHE policy evaluation from seconds to milliseconds, making real-time cross-domain filtering practical.

## 4.4 EVM Precompile Interface

Smart contracts can invoke FHE operations via precompiles:

| Address | Function | Gas |
|---|---|---|
| `0x0700` | FHE encrypt/decrypt/eval | 500,000 |
| `0x0701` | Ciphertext ACL management | 25,000 |
| `0x0702` | Input proof verification | 50,000 |
| `0x0703` | Cross-chain gateway | 100,000 |

Table 3: FHE EVM precompiles for on-chain policy enforcement.

# 5 Key Management for Multi-Level Security

## 5.1 Hanzo KMS Architecture

The KMS provides 8 subsystems supporting MLS key lifecycle:

1. **Shamir Secret Sharing**: Root key split into $n$ shares over $\mathrm{GF}(2^8)$; $t$ required for reconstruction. Custodians geographically distributed.
2. **Transit Engine**: Encryption-as-a-service with key versioning. Each security domain has independent transit keys with separate rotation schedules.
3. **HPKE Key Wrapping**: Content encryption keys (CEKs) wrapped using hybrid HPKE (X25519 + ML-KEM-768) per RFC 9180.
4. **Seal/Unseal Barrier**: AES-256-GCM barrier protecting all stored secrets. Auto-unseal via AWS/GCP KMS for operational environments.
5. **ACL Policy Engine**: Path-based capabilities with template variables (`{{identity.org_id}}`) enabling per-domain access rules.
6. **Token Hierarchy**: Service, batch, and recovery tokens with parent-child revocation cascade. Compromised token revocation propagates instantly.
7. **Dynamic Secrets**: Temporary database credentials (PostgreSQL, MySQL, Redis, MongoDB) scoped to security domain and auto-expired.
8. **TFHE-KMS Bridge**: FHE key management integrated with policy evaluation pipeline.

## 5.2 Cross-Domain Key Isolation

- Each security domain has an independent root key, never shared.
- Cross-domain data transfer requires re-encryption under destination domain key.
- Re-encryption occurs only after FHE policy approval and threshold decryption.
- No single KMS instance holds keys for multiple domains.

## 5.3 Compliance Controls

| Standard | Implementation |
|---|---|
| HIPAA | Break-glass tokens (time-limited), WORM audit logs |
| SEC 17a-4 | 6–7 year retention, immutable SHA-256 chain |
| SOX | Regulator escrow shards, cooperative reconstruction |
| GDPR | Per-org keys, graceful revocation |
| CNSA 2.0 | PQ algorithms for all key operations |

Table 4: Regulatory compliance controls in KMS.

# 6 Audit and Accountability

## 6.1 Blockchain-Anchored Audit Trail

Every cross-domain transfer produces an immutable audit record:

- **Record contents**: Source domain, destination domain, policy ID, decision (approve/deny), timestamp, data hash (not data), requestor DID.
- **Signature**: ML-DSA-65 (post-quantum) ensuring long-term non-repudiation.
- **Anchoring**: Record hash committed to Lux blockchain via Quasar consensus.
- **WORM guarantee**: Write-once-read-many with SHA-256 hash chain prevents retroactive modification.

## 6.2 Attestation Protocol

Domain-typed attestations with equivocation detection:

- **CrossDomainTransfer**: Records each approved transfer with full provenance.
- **PolicyUpdate**: Records changes to cross-domain policies.
- **KeyRotation**: Records domain key rotation events.
- Signing contradictory attestations (equivocation) is automatically detected and flagged.

# 7 Bell-LaPadula Enforcement

## 7.1 Mandatory Access Control

The architecture enforces Bell-LaPadula properties:

**Definition 7.1** (Simple Security Property (No Read Up))**.** A subject at security level $L_s$ cannot read an object at security level $L_o$ where $L_o > L_s$. Enforced by the zero-trust fabric: identity JWT contains org-scoped level; gateway rejects requests to higher-level services.

**Definition 7.2** (*-Property (No Write Down))**.** A subject at security level $L_s$ cannot write to an object at security level $L_o$ where $L_o < L_s$. Enforced by the FHE guard: data flowing from high to low domain must pass policy evaluation.

## 7.2 Implementation

1. **Read control**: Gateway checks `X-Org-Id` (derived from JWT) against service security label. Requests to higher domains rejected before reaching backend.
2. **Write control**: All data egress from a domain passes through the FHE guard. Downward flows require policy approval; upward flows are unrestricted.
3. **Label integrity**: Security labels are gateway-derived from cryptographic JWT claims. Client-supplied labels are stripped. Labels cannot be forged without compromising the IAM signing key.

# 8 Comparison with Existing CDS Approaches

| **Feature** | **Traditional Guard** | **Data Diode** | **FHE Guard (Ours)** |
|---|---|---|---|
| Plaintext exposure | Guard sees all | One-way only | Never |
| Bidirectional | Yes | No | Yes |
| Content inspection | Full | None | On ciphertext |
| Single point of trust | Guard | No | No (threshold) |
| PQ-safe | Depends | N/A | Yes (FIPS) |
| Audit | Guard logs | Physical | Blockchain |
| ML classification | On plaintext | N/A | On ciphertext |

Table 5: CDS approach comparison.

The FHE-based approach uniquely provides bidirectional content-inspected cross-domain transfer with zero plaintext exposure. No commercially available CDS offers this capability.

# 9 Performance Considerations

## 9.1 FHE Overhead

| **Operation** | **CPU** | **GPU (Metal)** |
|---|---|---|
| TFHE gate evaluation | 13 ms | 0.8 ms |
| BFV keyword match | 45 ms | 3.2 ms |
| CKKS ML inference | 200 ms | 15 ms |
| Threshold decrypt ($t$=3) | 50 ms | 12 ms |
| **Total policy eval** | **∼300 ms** | **∼30 ms** |

Table 6: FHE policy evaluation latency.

GPU acceleration brings FHE policy evaluation to 30 ms—practical for real-time cross-domain filtering of message traffic, though not suitable for bulk file transfer without parallelization.

## 9.2 Throughput

With GPU acceleration and pipelined evaluation:

- **Messages**: ∼33 messages/second per GPU (real-time chat feasible).
- **Bulk data**: Parallelize across multiple GPUs; 8-GPU node achieves ∼264 evaluations/second.
- **Ciphertext expansion**: 100–1000× plaintext size (FHE-inherent). ZAP zero-copy design mitigates the internal memory cost.

# 10 Naval MLS Deployment Scenarios

## 10.1 Coalition Information Sharing

- US Secret data evaluated against release policy for Five Eyes partners.

- FHE guard checks classification markers, source restrictions, and content sensitivity—all on ciphertext.
- Approved data re-encrypted under coalition domain key and forwarded.
- Full audit trail anchored to blockchain for post-operation review.

### 10.2 Analyst Workstation Cross-Domain Access

- Analyst at Secret level queries Top Secret/SCI database.
- Query encrypted under FHE key; database returns encrypted results.
- FHE guard evaluates sanitization policy on encrypted response.
- Approved (sanitized) response decrypted for analyst's domain.

### 10.3 Shipboard MLS Network

- Multiple security domains on single shipboard network (physically separated VLANs, logically separated via zero-trust overlay).
- FHE guards at domain boundaries, GPU-accelerated for real-time messaging.
- KMS with separate root keys per domain, HSM-protected.
- All cross-domain traffic audited with PQ-signed blockchain receipts.

## 11 Conclusion

We have presented a cross-domain solution architecture where the guard evaluates policies on encrypted content, eliminating the traditional single point of trust. The combination of FHE policy evaluation, zero-trust network fabric, threshold MPC decryption, and blockchain-anchored audit provides a multi-level security architecture with no plaintext exposure at the guard, no single point of key compromise, and immutable accountability.

This approach represents a paradigm shift in cross-domain security: instead of trusting the guard to handle plaintext responsibly, we make it cryptographically impossible for the guard to access plaintext at all.

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